Unsupervised Domain Adaptation for SAR Target Detection

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***Abstract*—Recent years have witnessed great progress in synthetic aperture radar (SAR) target detection methods based on deep learning. However, these methods generally assume the training data and test data obey the same distribution, which does not always hold when the radar parameters, imaging algorithm, viewpoints, scenes, etc., change in practice. When such a distribution mismatch occurs, it will cause a significant performance drop. Domain adaptation methods provide an effective way to address this problem by transferring knowledge from the source domain (training data) to the target domain (test data). In this article, we proposed an unsupervised faster R-CNN SAR target detection framework based on domain adaptation, which can improve SAR target detection performance in the unlabeled target domain by borrowing the knowledge of the labeled source domain. Our approach is composed of the following three stages: pixel-domain adaptation (PDA), multilevel feature domain adaptation (MFDA), and iterative pseudolabeling (IPL). By generating transition domain using generative adversarial networks, the PDA stage can reduce the appearance differences of SAR images. At the MFDA stage, the detector can not only learn the domain-invariant global features and instance-level regional features via multilevel adversarial learning in the common feature space but also reweight the low-level global features according to their relative importance to the target domain. At the IPL stage, we design an iterative pseudo labeling strategy that can select pseudo-labels on instance level and image level to encourage the detector to learn more discriminative features of the target domain directly. We evaluate our method using miniSAR and FARADSAR datasets. The experimental results demonstrate the effectiveness of the proposed unsupervised domain adaptation target detection approach.**

***Index Terms*—Adversarial learning, iterative pseudo labeling (IPL), synthetic aperture radar (SAR), target detection, unsupervised domain adaptation.**

I. Introduction

With the advantage of providing remote sensing images under all-day and all-weather conditions, synthetic aperture radar (SAR) is widely used in the military and civilian fields. Over the past several years, the advancement in SAR-related technologies, which makes it possible to acquire a large amount of higher resolution SAR images, has significantly pushed forward the development of SAR image interpretation. As a fundamental and challenging task in SAR image interpretation, SAR target detection is obtaining wide attention [2]–[8].

With the recent development of the deep learning, a large number of methods based on convolutional neural networks (CNNs) are proposed. Due to numerous training data that can be learned by the network, these methods have achieved significant progress in target detection, among which the faster R-CNN [1] is one of the most widely used methods and can be extended to other tasks flexibly. Inspired by great success in target detection of optical images, the use of CNNs in SAR target detection [2]–[8] is also obtained wide attention. Kang *et al.* [4] proposed an improved faster R-CNN by fusing features of the last three convolution layers. Jiao *et al.* [5] connected multiscale feature map densely based on faster R-CNN to solve multiscale and multiscene problems.

While excellent performance has been achieved, SAR target detection methods based on CNNs still face great challenges from the large variance in the radar parameters, imaging algorithm, viewpoints, scenes, etc., which may cause a distribution mismatch of the training data and test data in practice. Such mismatch will cause a significant performance drop. A feasible way is to collect more labeled SAR images to train a more generalized detector for the target domain (test data), but annotating bounding boxes for SAR images with complex scenes is expensive and time-consuming. Therefore, it is worth developing algorithms to adapt an SAR target detector to a new domain whose distribution is different from the training domain. Domain adaptation methods provide an effective way to address this problem by transferring knowledge from the fully labeled source domain to the target domain with unlabeled or insufficient-labeled data. In the area of computer vision, domain adaptation methods for an image classification task have already been widely studied. These domain adaptation methods use the maximum mean discrepancy (MMD) [9]–[11], gradient reversal layer (GRL) [12]–[15], or translating images from the source domain to the target domain [16]–[18] to reduce domain discrepancy. In the area of SAR target recognition, Huang *et al.* [19] explored what, where, and how to transfer knowledge from other domains to the SAR domain through MMD constraint.

Compared with the domain adaptation methods for classification, there is limited research on domain adaptation for target detection since target detection is a more challenging task, which predicts not only the category but also the location. Inoue *et al.* [20] achieved a two-step weakly supervised domain adaptation framework by using conventional pixel-level domain

Manuscript received March 15, 2021; revised May 23, 2021; accepted June 4, 2021. Date of publication June 14, 2021; date of current version July 14, 2021. This work was supported in part by the National Science Foundation of China under Grant 61771362 and in part by the 111 Project under Grant B18039. *(Corresponding author: Lan Du.)*

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Digital Object Identifier 10.1109/JSTARS.2021.3089238

adaptation methods and pseudo labeling. Chen *et al.* [21] and Saito *et al.* [22] learned domain-invariant features by aligning feature distribution through adversarial learning. He and Zhang [23] and Chen *et al.* [24] proposed an importance-weighted GRL to reweight source or interpolation samples. Kim *et al.* [25] solved domain adaptation by utilizing multiple GANs with color retention and other constraints to generate multiple intermediate domains to reduce domain discrepancy and using multidomain-invariant representation learning to alleviate the imperfect translation problem. The abovementioned methods can solve the lack of labeled data problem in target detection tasks for optical images through domain adaptation. However, most of the abovementioned methods only indirectly learn the common features of the target domain and the source domain by narrowing the distance between the two domains, which lose some more discriminative features of the target domain. Besides that, it will lead to a performance drop when the abovementioned methods are directly used in SAR images due to the following reasons. First, the SAR images mainly include structure and shape information without color information, which makes the methods involving color transformation unable to learn image features effectively. Second, compared with sufficient optical data in the source domain, the number of SAR images in the source domain is relatively small, which limits the application of the abovementioned methods. In this article, we focus on unsupervised domain adaptation for SAR target detection, where the source domain data is fully labeled while the target domain data is unlabeled. To the best of our knowledge, there is no relevant study on SAR target detection yet.

In this article, a novel unsupervised domain adaptation framework for SAR target detection based on faster R-CNN is proposed, which can improve the detection performance of the target domain without expensive label cost. The proposed method consists of three stages: pixel domain adaptation (PDA), multilevel feature domain adaptation (MFDA), and iterative pseudo labeling (IPL). At the PDA stage, CycleGAN [26] is used to generate the transition domain to reduce the appearance differences of SAR images. In particular, we combine the source domain and the transition domain to form a new domain, named diverse domain, which increases the number of trainable data and enriches the distribution of the labeled data. MFDA aims to align feature distribution at multilevel common feature space through adversarial learning. First, the detector can learn the domain-invariant low- and high-level global features and instance-level regional features through adversarial learning in the common feature space. Second, as not all samples are equally treated, especially in the transition domain with the imperfect translation problem [25], the low-level global features are reweighted according to their relative importance to the target domain. PDA and MFDA can help the detector transfer knowledge from the source domain to the target domain, but it will lead to the domain-biased problem because the target domain is unlabeled while the diverse domain is labeled. At the IPL stage, we design an IPL strategy that can select pseudo-labels on instance level and image level to solve this problem by encouraging the detector to learn more discriminative features of the target domain directly.

Our method has obvious differences from existing methods. Most existing methods only indirectly learn the common features of the target domain and the source domain by narrowing the distance between the two domains, which lose some of the more discriminative features of the target domain. Although Inoue *et al.* [20] use pseudo labeling to learn the feature of the target domain directly, it is a weakly supervised framework that only focuses on instance-level pseudo-labels and ignores the quality of image-level pseudo-labels. Compared with the GAN-based methods, in addition to the differences in the network architecture, we also consider the limited training samples and SAR images that mainly contain structure, texture, and shape features without colors. Our method generates the transition domain to enrich the distribution of the labeled data in the PDA stage and reweights the features at the low-level feature space according to their similarity in the MFDA stage. More specifically, the main contributions of the proposed method can be summarized as follows.

1) It is the first work for unsupervised domain adaptation SAR target detection, which can improve the SAR target detection performance of the target domain without expensive label cost.
2) We proposed IPL that can select high-quality pseudo-labels on instance level and image level to solve the domain-biased problem.
3) PDA and MFDA are improved to alleviate domain discrepancy from pixel and feature aspects.

We evaluate our method using miniSAR and FARADSAR measured datasets. The experimental results clearly demonstrate the effectiveness of the proposed unsupervised domain adaptation target detection approach.

The rest of this article is organized as follows. Section II introduces the proposed novel unsupervised domain adaptation framework for SAR target detection. Section III gives the experimental results and corresponding analysis based on the miniSAR and FARADSAR measured datasets. Finally, Section IV concludes this article.

II. Proposed Target Detection Method

In this section, we will introduce the details of the proposed method. Formally, a source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ with N_s fully labeled SAR images, where y_i^s stands for the groundtruth annotations of x_i^s, and a target domain $\mathcal{D}_t = \{(x_j^t)\}_{j=1}^{N_t}$ with N_t fully unlabeled SAR images are available. Without loss of generality, the source domain and the target domain have different distributions. Therefore, when directly applying the detector trained on the source domain to the target domain, it will lead to a significant performance drop. The purpose of our method is to improve the detection performance of the target domain without any expensive label cost by borrowing the knowledge of the source domain.

The overview of the proposed unsupervised domain adaptation framework for SAR target detection is shown in Fig. 1, which consists of three stages: PDA, MFDA, and IPL. The framework is based on faster R-CNN target detector and uses VGG16 as the backbone network. The PDA stage applies GANs

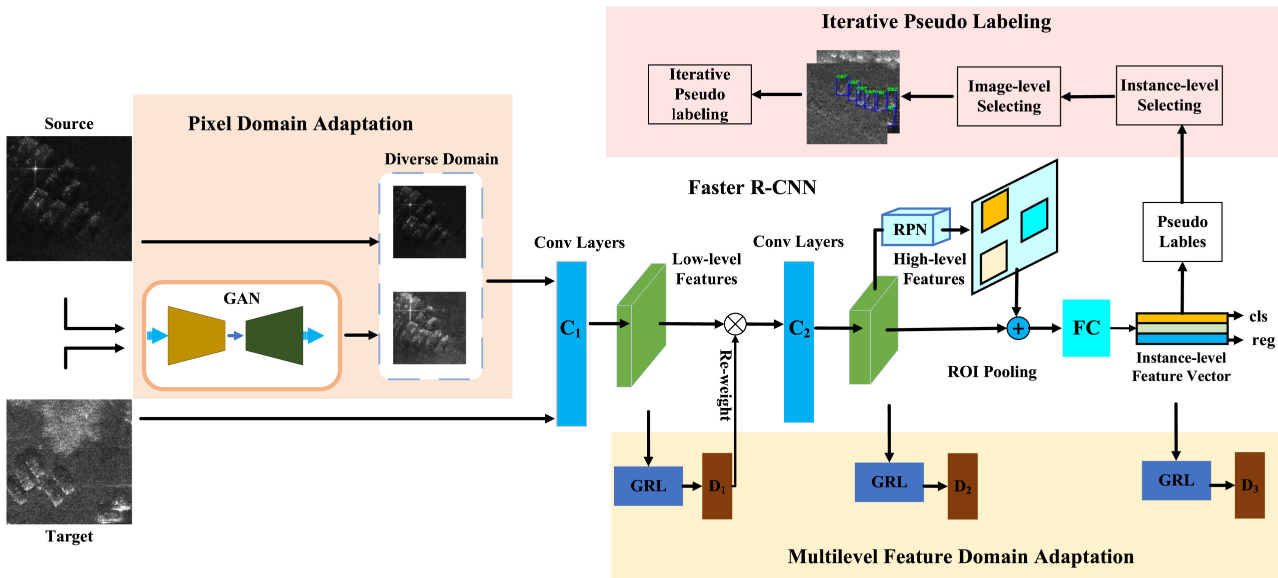

Fig. 1. Architecture of unsupervised domain adaptation for SAR images target detection. C_1 and C_2 denote low-level and high-level feature extractors, whereas FC denotes the fully connected layers. D_1 and D_2 are global domain classifiers, whereas D_3 is the instance domain classifier.

to generate the transition domain and combine them to form the diverse domain to enrich the distributions of the labeled data. MFDA not only learns the domain-invariant low-level global features, high-level global features, and instance-level regional features via multilevel adversarial learning in the common feature space but also reweights the low-level global features according to their relative importance. At the IPL stage, we design an IPL strategy that can select pseudo-labels on instance level and image level to encourage the detector to learn more discriminative features of the target domain directly.

In the following sections, the structure of each stage is introduced in detail.

A. Pixel Domain Adaptation (PDA)

Due to the changes in radar parameters, imaging algorithm, viewpoints, scenes, etc., the SAR images in different SAR domains generally have different appearances. Although some traditional methods based on image grayscale adjustment can change the brightness of SAR images, they need different adjustments for different SAR images and cannot learn the distribution of the target domain to fill the distribution gap. To alleviate such a domain discrepancy, CycleGAN, an unsupervised image-to-image translation model of GANs, is used to generate transition domain by learning the distributions of the target domain. The CycleGAN is trained to generate samples that have similar appearances with the target domain, whereas the location and pose of the targets of interest (e.g., vehicles) remain unaltered, which means that they can share common labels with the source domain samples. The motivation of the PDA is illustrated in Fig. 2. Ideally, the transition domain has the same distribution as the target domain. However, due to the difficulty in optimization and mode collapse of GANs, the imperfect translation problem [25] cannot be ignored, making the gap between the transition domain and the target domain still

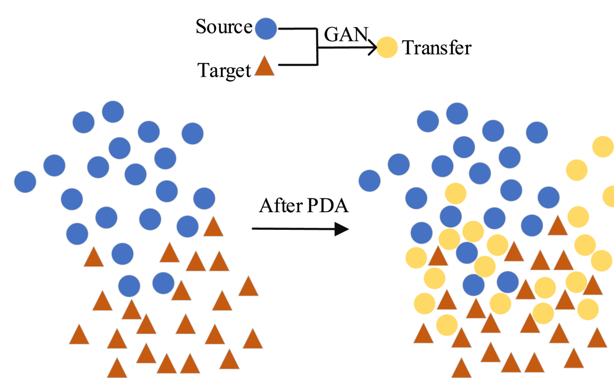

Fig. 2. Motivation of the PDA for reducing the differences between the source domain (blue points) and the target domain (red triangles) by generating the transition domain (orange points).

exists. The transition domain can be thought of as the domain that moves along the source domain to the target domain. To mitigate the imperfect translation problem and meanwhile solve the fewer training samples problem, we combine the source domain and the transition domain to form a new domain, named diverse domain, which can increase the number of trainable data and enhance the generalization ability of the model by enriching the distribution of the labeled data.

B. Multilevel Feature Domain Adaptation (MFDA)

Domain adversarial learning, such as GRL [12], is a typical domain alignment approach to align feature distributions via an adversarial learning process. The MFDA stage aims to align feature distributions between the diverse domain and the target domain at multilevel common feature space through adversarial learning alignment. In a deep target detection network, global features reflect the image information (such as contour, edge, texture, and shape) and instance-level features reflect the target

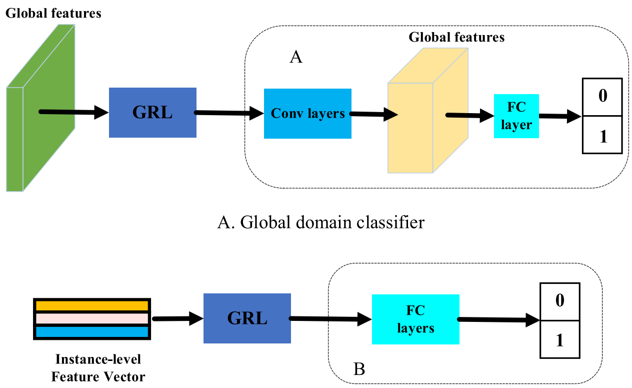

Fig. 3. Structure of the domain classifier. A is the domain classifier for low- and high-level global features, whereas B is the domain classifier for instance-level regional features. "0" and "1" denote predicted domain labels.

information (such as target appearance and target viewpoint). Therefore, MFDA is proposed to alleviate domain discrepancy via multilevel adversarial learning.

First, we aim to align low- and high-level global features to reduce global image differences and align instance-level regional features to reduce regional target differences. The abovementioned process is implemented by GRL and domain classifiers. As shown in Fig. 3, we apply different classifiers for global features and instance-level regional features. Given an image x_i with a domain label d_i, where $d_i = 0$ denotes the image from the diverse domain and $d_i = 1$ denotes the image from the target domain. The low-level global features are extracted by the middle convolution layer in the base network (C_1 in Fig. 1) and the high-level global features are extracted by the last convolution layer in the base network (C_2 in Fig. 1). The domain classifier $D_k, k \in \{1, 2\}$ is trained to predict the domain label of input global features. To align the domain distributions, the domain classifier aims to distinguish the features of two domains and the feature extractor (the convolution layers) aims to confuse the domain classifier. This adversarial process is achieved through GRL, and the domain classification loss for low- and high-level global features can be written as:

$$\mathcal{L}_g = -\sum_{i,k} [d_i \log p_i^k + (1 - d_i) \log(1 - p_i^k)] \quad (1)$$

where p_i^k denotes the output of the domain classifier D_k, which p_i^k is the probability of the image x_i belonging to the diverse domain, and $(1 - p_i^k)$ is the probability of the image x_i belonging to the target domain. The parameters of the domain classifier should be optimized to minimize the above loss and the parameters of the feature extractor should be optimized to maximize this loss. After the adversarial learning of the domain classifier, the feature extractor can extract the domain-invariant low- and high-level global features.

The instance-level regional features are represented as the ROI-based feature vectors after the fully connected layer. Similar to the global feature alignment, the domain classifier D_3 is trained to align the instance-level regional features distribution to reduce target differences. The domain classification loss for instance-level regional features can be written as

$$\mathcal{L}_{\text{ins}} = -\sum_{i,j} [d_i \log p_{i,j} + (1 - d_i) \log(1 - p_{i,j})] \quad (2)$$

where $p_{i,j}$ denotes the output of the domain classifier D_3 for the jth proposal in the image x_i. The same adversarial strategy is achieved through GRL.

Second, we aim to reweight the labeled samples at the low-level global feature space according to their relative importance to the target domain. A reasonable assumption is that the closer the distribution to the target domain, the higher importance of the sample is. Therefore, not all samples should be equally treated, especially in the transition domain with the imperfect translation problem. A weighted GRL [25], which regards the outputs of the domain classifier as the weights for the corresponding samples, is applied to achieve this process. Considering that SAR image features mainly include structure, texture and shape features without colors, we reweight features at the low-level feature space.

p_i^1 denotes the output of the domain classifier D_1 for sample x_i , and the corresponding weight w_i is measured by

$$w_i = -[p_i^1 \log(p_i^1) + (1 - p_i^1) \log(1 - p_i^1)]. \quad (3)$$

The higher weights mean that the samples are harder to be distinguished and should be upweighted. Otherwise, those samples with lower weights are considered to be easier to be distinguished and should be downweighted. The reweighted global features can be represented as

$$R_i = f_i + f_i \times w_i \quad (4)$$

where f_i denotes the low-level global features.

$\mathcal{L}_{\text{loc}}$and $\mathcal{L}_{\text{cls}}$ are the loss used to measure the accuracy of classification and position in the detection loss. The final training loss of the MFDA with combining the detection loss and the align loss can be written as

$$\mathcal{L} = \mathcal{L}_{\text{loc}} + \mathcal{L}_{\text{cls}} + \lambda(\mathcal{L}_g + \mathcal{L}_{\text{ins}}) \quad (5)$$

where λ is the parameter to balance the detection loss and the align loss.

C. *Iterative Pseudo Labeling (IPL)*

PDA and MFDA can help the detector transfer knowledge from the source domain to the target domain, but it will lead to the domain-biased problem because the target domain is unlabeled while the diverse domain is labeled. In other words, as the bonding box annotation is only obtained for the diverse domain, the detector with backpropagation is directed by diverse domain samples, which leads the detector easily to bias toward the diverse domain. Pseudo labeling strategy provides an effective way to solve such an issue. The samples in the diverse domain have accurate labels but cannot effectively learn the features of the target domain. On the contrary, although the samples in the target domain with pseudo-labels do not have completely correct annotations, more discriminative features can be directly learned as images are completely from the target domain. We proposed an IPL strategy that can select pseudo-labels on instance level

Algorithm 1: The Procedure of the Pseudo-Labels Selecting Methods.

Input: $L = \{x_i, b_{i,j}, p_{i,j}\}_{i=1}^{N}$ - the set of entire target domain training samples with pseudo-labels, where $b_{i,j}$ denotes the jth bounding box of ith image and $p_{i,j}$ denotes the probability of jth bounding box of the ith image; $conf$ - selection threshold of instance-level selecting;
Output: l - the subset of target domain training samples with selected pseudo-labels;
Initialization:
1. Set l as an empty set;
2. Set L_{ins} as an empty set;

Selection strategy:

3. if $p_{i,j} \geq conf$, select $(x_i, b_{i,j}, p_{i,j})$ into L_{ins};
4. Calculate $score_i = \frac{1}{n_i} \sum_{j=1}^{n_i} p_{i,j}$. for each image x_i containing n_i bounding boxes in L_{ins}, and sort by $score_i$ from high to low;
5. If $score_i$ is in the top k of the sorted sequence, set $p_{i,j} = 1$, then put $(x_i, b_{i,j}, p_{i,j})$ into l;
6. **return** l

and image level. For instance-level selection, a threshold is set to screen out less confident proposal bounding boxes. However, the instance-level pseudo-labels only focus on each proposal bounding box and ignore the overall pseudo labeling quality of the image, which is unreasonable. Consider the following situations: for some samples, setting a high selection threshold will produce many missed detections, whereas setting a low selection threshold will produce many false alarms. Such samples participating in the training will have a negative impact on the network. To alleviate the negative impact of false pseudo-labels on network training, we design a pseudo-labels selection criterion that can select pseudo-labels on instance level and image level. The selection criterion is shown in Algorithm 1. The selected samples participate in the first iterative training of the faster R-CNN network, and the remaining samples are pseudo-labeled for the second time by the trained faster R-CNN network:

$$y_j = \begin{cases} 1, p_j \geq \text{conf.} \\ 0, \text{Otherwise.} \end{cases} \tag{6}$$

Then, we combine the samples annotated the second time with the samples selected the first time to participate in the second iterative training.

III. Experimental Results and Analysis

A. Description of the Dataset and Experimental Settings

The proposed method is verified with the measured miniSAR and FARADSAR datasets, which are acquired by U.S. Sandia National Laboratories in 2005 and 2015, respectively.[1] The image size in the miniSAR dataset is 1638×2510, and the image size in the FARADSAR dataset is between 1300×580 and 1700×1850. Nine SAR images in the miniSAR dataset are selected for the experiment, seven of which are used for training and two for testing, and one hundred six SAR images in the FARADSAR dataset are selected for the experiment, seventy eight of which are used for training and twenty eight for testing. The miniSAR dataset and the FARADSAR dataset are acquired from two different radars. Table I shows the acquisition information of the two datasets [8], [27]–[29], from which we can see that they have a large distribution gap between the two datasets. The characteristic of SAR images will change greatly with different operating conditions, including configuration, grazing angle, aspect view, and clutter [30], [31]. Since the grazing angle and band are different, the backscattering characteristic of the target is different. Since the scenes are different (one in the suburbs and one with dense buildings), the clutter distribution is different. Fig. 4 shows two SAR images of miniSAR dataset and FARADSAR dataset and two corresponding optical images. From Fig. 4, we can see that the appearances of SAR images with different datasets have a large domain gap and the scenes have a big difference. Compared with miniSAR dataset, FARADSAR dataset has more scenes and stronger clutter.

TABLE I
Acquisition Information of the miniSAR Data and the FARADSAR Data

System Characteristics	miniSAR	FARADSAR
Mode	Spotlight	Stripmap
Band	Ku	Ka
Center Frequency	16.8GHz	35.6GHz
Grazing Angle	26°-29°	29°-34°
Location	Kirtland Air Force Base	The University of New Mexico
Scenes	Golf Course, Helicopter Park, Baseball Field, etc.	Dense urban buildings
Time	2005	2015
Resolution	4-inch	4-inch

For the image size, the original SAR images are not suitable for being the input of the network due to the large size. Therefore, the original SAR images are cropped into many subimages with a fixed size of 300×300, then we got 143 miniSAR images with 800 vehicles and 574 FARADSAR images with 2157 vehicles. For the testing process, the original test SAR images are also cropped into many test subimages with the size of 300×300 by sliding window repeatedly. Then, all the test SAR subimages are input to the target detection network, and the prediction results of all subimages are restored to the original SAR image. Finally, the non-maximum suppression (NMS) deduplication algorithm is employed to obtain the final detection results.

The established domain adaptation tasks include: first, from the miniSAR dataset to the FARADSAR dataset (M2F); and from the FARADSAR dataset to the miniSAR dataset (F2M).

[1] [Online]. Available: https://www.sandia.gov/radar/complex-data/

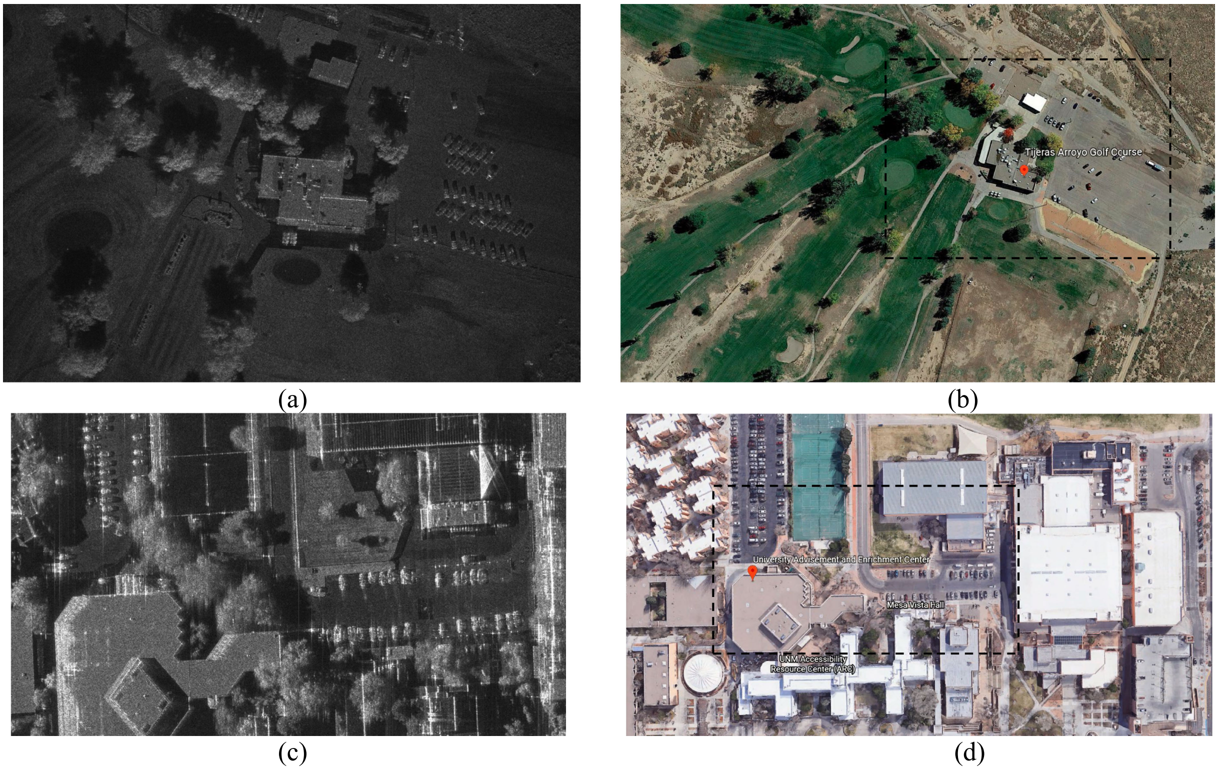

Fig. 4. Examples of miniSAR dataset and FARADSAR dataset. (a) Image of miniSAR dataset. (b) Corresponding optical scene to (a) on Google Earth: Tijeras Arroyo Golf Course in Kirtland Air Force Base. (c) Image of FARADSAR dataset. (d) Corresponding optical scene to (c) on Google Earth: Advisement and Enrichment Center in the University of New Mexico.

For the M2F task, all training images of miniSAR are utilized as the source domain, all training images of FARADSAR are utilized as the target domain, and 28 testing images of FARADSAR are utilized to test. For the F2M task, all training images of FARADSAR are utilized as the source domain, all training images of miniSAR are utilized as the target domain and 2 testing images of miniSAR are utilized to test.

In our experiment, stochastic gradient descent with a momentum of 0.9 is used for the training procedure and the initial learning rate is set to 0.001. We set λ=0.15 in (5) for the IPL, and the threshold is set to 0.5 in (6) and Algorithm I. k is set to half of all training samples in the target domain in Algorithm I.

The experiments are conducted using a personal computer with Intel Xeon E5-2620 v4 CPU of 2.1 GHz, NVIDIA GeForce GTX 2080Ti GPU, 128 GB of memory on Ubuntu 18.04 Linux system and implemented with the Pytorch deep learning framework.

B. Evaluation Criteria

We use precision, recall, and F1-score to evaluate different detection methods quantitatively. The calculation is formulated as follows:

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{8}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{9}$$

where TP represents the number of correctly detected vehicle targets, FP represents the number of false alarms, and FN represents the number of missed detections. The precision and the recall, respectively, measure the fraction of true positives overall detected results and the fraction of true positives over the ground truths. In other words, the higher the precision and recall, the lower the false alarm rate and missed detection rate. The F1-score, which is the harmonic mean between precision and recall, can comprehensively evaluate the performance of target detection.

C. Comparison With Other Methods

To demonstrate the excellent performance of the proposed method, we compare it with two famous unsupervised domain adaptation target detection methods. Table II and Figs. 5 and 6 show the target detection results of the proposed method and other detection methods on two test SAR images in miniSAR dataset and FARADSAR dataset. In Table II and Figs. 5 and 6, the baseline only stands for the detection network that is trained only using source images without adaptation. Domain adaptive faster R-CNN (DAF) and hierarchical transferability calibration network (HTCN), respectively, denote the unsupervised domain adaptation method in [21] and [24]. The upper

TABLE II
OVERALL EVALUATION OF DIFFERENT UNSUPERVISED TARGET DETECTION METHODS

	M2F			F2M		
	Precision	Recall	F1-score	Precision	Recall	F1-score
Baseline	0.4491	0.8133	0.5787	0.9524	0.6504	0.7729
DAF	0.4823	0.8640	0.6191	0.9053	0.7073	0.7941
HTCN	0.5182	0.8365	0.6400	0.8788	0.7073	0.7838
Proposed method	0.7968	0.8090	**0.8028**	0.8595	0.8455	**0.8525**
Upper bound	0.8436	0.9132	0.8770	0.8421	0.8943	0.8674

bound denotes the fully supervised faster R-CNN detection method.

For quantitative analysis, we evaluate the overall target detection results on 2 test SAR images in the miniSAR dataset and 28 test SAR images in the FARADSAR dataset in terms of precision, recall, and F1-score in Table II. For the F2M task, as the FARADSAR dataset has more data and scenes, the baseline that only uses source images without adaptation still achieves guaranteed results. The M2F task is the opposite. In particular, compared with the baseline, the proposed method achieves +22.41% F1-score performance in the M2F task, but it is still lower than the upper bound by 7.42% since the M2F task is a more challenging task that transfers knowledge from the domain with fewer data and scenes to the domain with more data and scenes. For the F2M task, the proposed method achieves +7.96% performance, which is very close to the upper bound (only 1.49% apart). Compared with other domain adaptation methods, the proposed method beyond DAF and HTCN +18.37% and +16.28%, respectively, in the M2F task and beyond DAF and HTCN +5.84% and +6.87%, respectively, in the F2M task. This is because the proposed method designs a more suitable framework for SAR images and uses pseudo-labels to learn more discriminative features. The proposed method significantly outperforms all comparison methods, which demonstrates the effectiveness of our method.

Figs. 5 and 6 exhibit the intuitional target detection results of the proposed method and other methods on two test SAR images in miniSAR dataset and two test SAR images in FARADSAR dataset. For the domain adaptation detection results in M2F task, since the scenes of the miniSAR dataset are simpler and have less strong scattering clutter, the baseline, which is trained only using source images without adaptation, generates lots of missed detections when tested to FARADSAR data. Since the methods based on domain adaptation can learn the common features of the source domain and the target domain by adding constraints, they can reduce missed detections to varying degrees. In particular, the proposed method can learn more discriminative features of the target domain and outperforms all comparison methods. On the opposite, for the domain adaptation detection results in F2M task, since the scenes of FARADSAR dataset are more complex and have more strong scattering clutter, the baseline, which is trained only using source images without adaptation, generates a lot of false alarms when tested to miniSAR data. The methods based on domain adaptation reduced false alarms to varying degrees and the proposed method outperforms all comparison methods.

TABLE III
ABLATION STUDY ON M2F

	PDA	MFDA	IPL	Precision	Recall	F1-score
Source only				0.4491	0.8133	0.5787
Ours	✓			0.5567	0.8553	0.6744
		✓		0.6509	0.6903	0.6700
			✓	0.4884	0.8596	0.6229
	✓	✓		0.7877	0.6831	0.7316
	✓		✓	0.6004	0.8538	0.7050
		✓	✓	0.6503	0.7829	0.7105
	✓	✓	✓	0.7968	0.8090	**0.8028**

D. Model Analysis

1) Ablation Study: To analyze the different stages of the proposed method, we designed and implemented ablation studies using the miniSAR dataset and the FARADSAR dataset. By analyzing the results of the ablation study, we can understand the impact of different modules on the detection performance more comprehensively. The results of the ablation study are shown in Tables III and IV, where PDA, MFDA, and IPL denote whether to use the pixel domain adaptation, multilevel feature domain adaptation, and iterative pseudo labeling, respectively.

From Tables III and IV, we can see that the PDA, the MFDA, the IPL can all improve the detection performance. Specifically, for the M2F task, the domain adaptation is a process to reduce false alarms, and for the F2M task, the domain adaptation is a process to reduce missed detections. But in this process, the increase in precision/recall is accompanied by varying degrees of decrease in recall/precision, especially the MFDA with learning domain-invariant features. IPL alleviates this problem

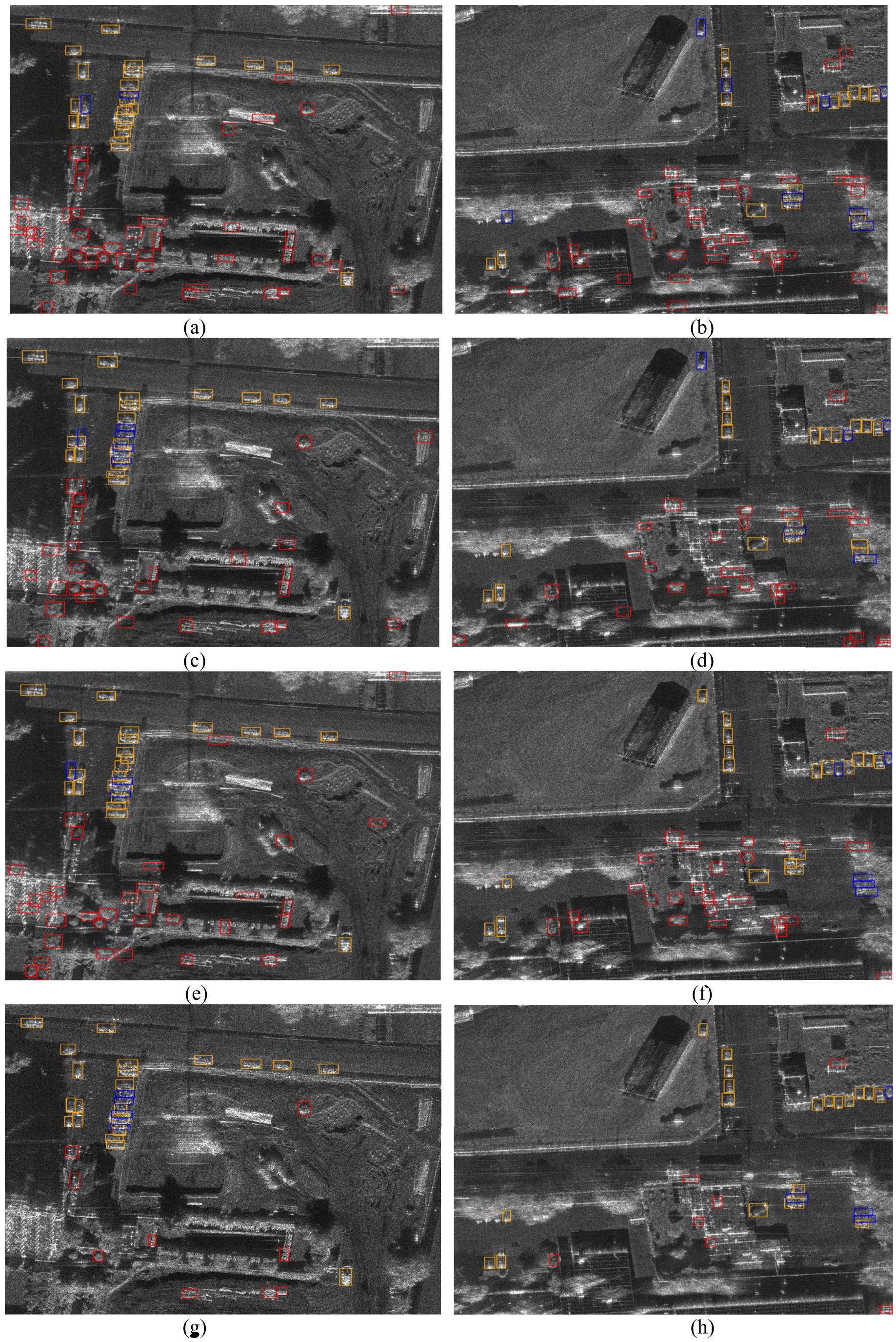

Fig. 5. Intuitional target detection results comparing with other methods on two test SAR images in FARADSAR, where yellow rectangular boxes represent the detected targets are correct, red rectangular boxes represent the detected targets are false alarms, and blue rectangular boxes represent the missed detections. (a) and (b) Baseline. (c) and (d) DAF. (e) and (f) HTCN. (g) and (h) Proposed method.

by learning more discriminative features directly. Using only PDA or MFDA achieves a similar F1-score improvement, and the detection performance of using IPL only is not obvious. According to the ablation study of the IPL stage, we can see that the better the detector performance, the better the performance of the pseudo labeling. This may be explained from the perspective of semisupervised self-training [32], [33]. The pseudo-labels obtained from the previous stage can be regarded as the initial pseudo-labels in self-training. Due to the problem of error accumulation [32], the performance of self-training often depends on the accuracy of pseudo-labels. IPL reduces the impact of error accumulation on network training by selecting pseudo-labels on instance level and image level. From Tables III and IV, we can conclude that the proposed method can achieve the best detection performance by using the PDA, MFDA, and IPL at the same time.

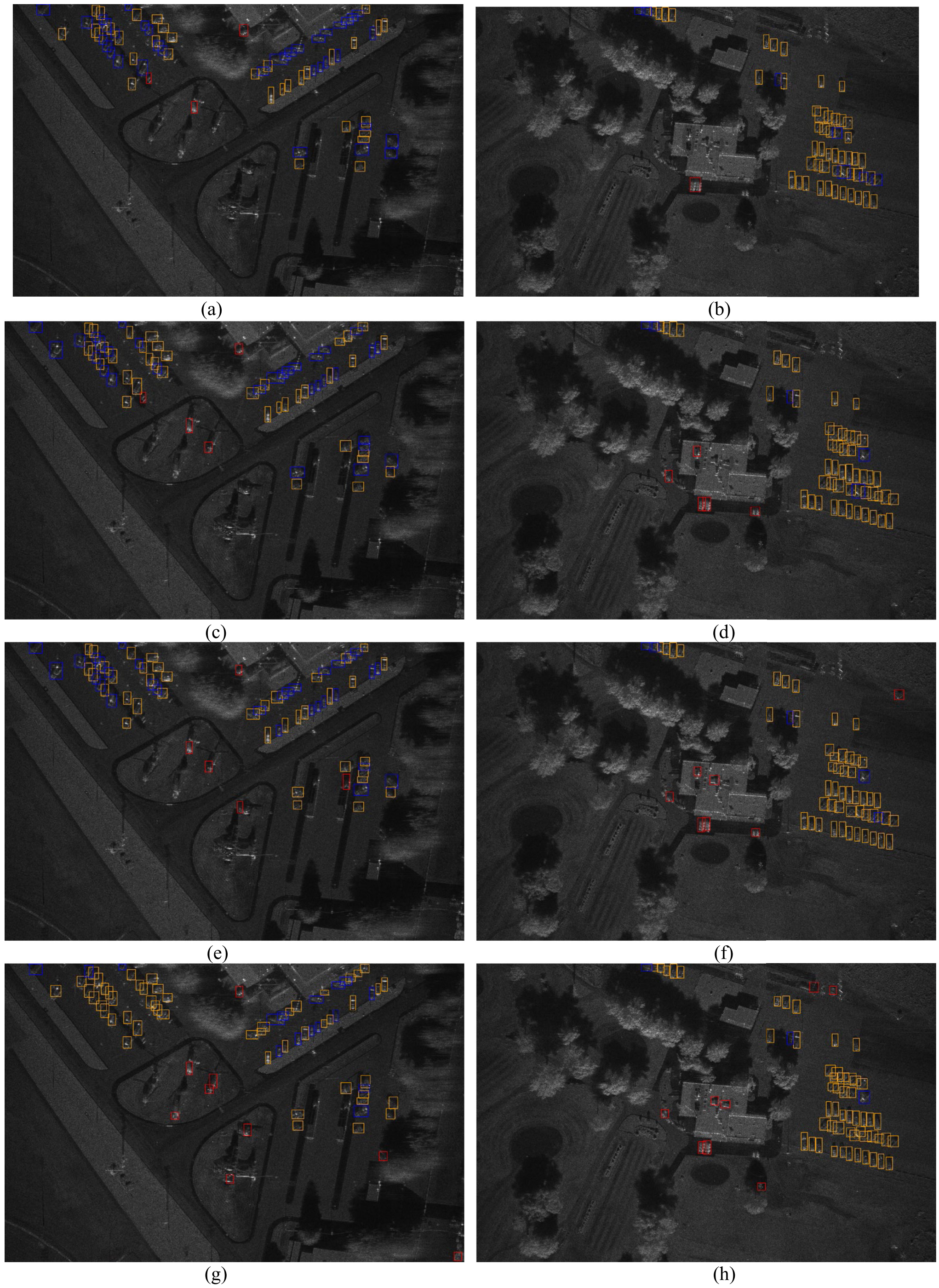

Fig. 6. Intuitional target detection results comparing with other methods on two test SAR images in miniSAR, where yellow rectangular boxes represent the detected targets are correct, red rectangular boxes represent the detected targets are false alarms, and blue rectangular boxes represent the missed detections. (a) and (b) Baseline. (c) and (d) DAF. (e) and (f) HTCN. (g) and (h) Proposed method.

2) Visualization of PDA: Qualitative results of the translations are presented in Figs. 7 and 8. As can be seen from the figures, the overall appearance of the images looks good, but there is still a difference with the target domain. Besides that, due to the difficulty of optimization and mode collapse, the translation quality of some images is not satisfactory, such as the last row of the middle part in Fig. 8. Combining the source domain and the transition domain to form the diverse domain can mitigate the imperfect translation problem and meanwhile increase the number of trainable data.

We further visualize the pixel-level image distributions before and after PDA by mapping the images to a low-dimensional 2-D space. Fig. 9 is the result of PDA from miniSAR (source domain) to FARADSAR (target domain), where each point represents the result of an image input to t-SNE. From this distribution plot, we can see that the transition domain (orange dot) moves along the source domain (blue dot) to the target domain (red triangle)

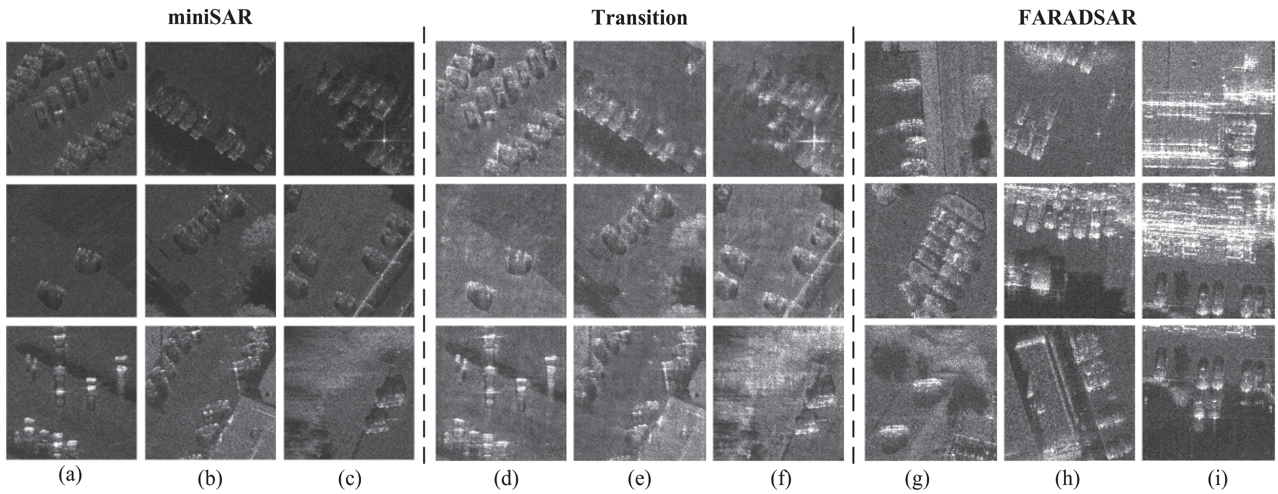

Fig. 7. CycleGAN transform results from miniSAR to FARADSAR, where the left part represents the SAR images in the miniSAR domain, the middle part represents the SAR images in the transition domain, and the right part represents the SAR images in FARADSAR domain. (a)–(i) Column of SAR images. Among them, the image in the transition domain corresponds to the transform result of images in miniSAR at the corresponding position.

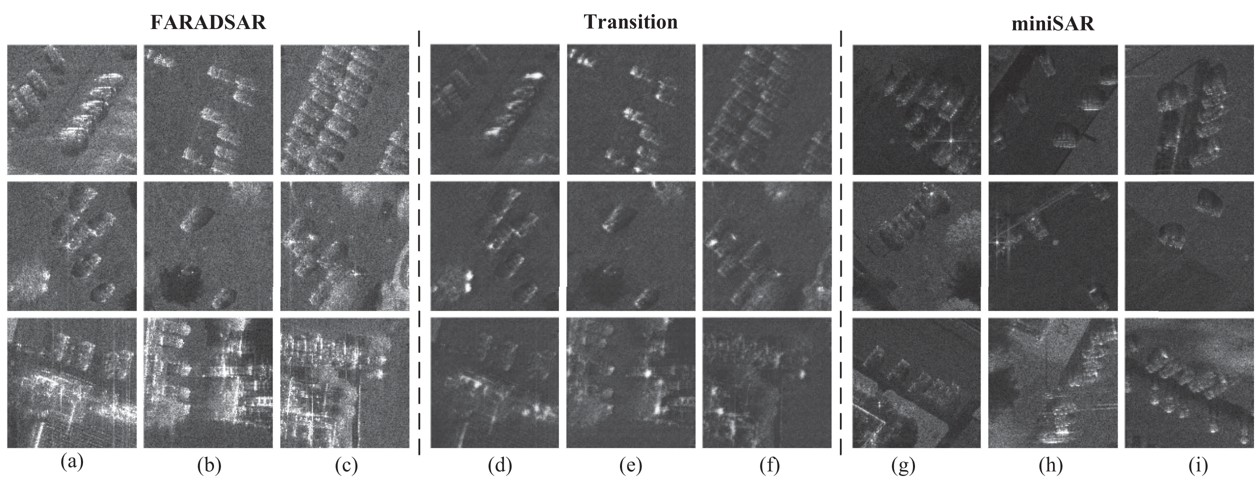

Fig. 8. CycleGAN transform results from FARADSAR to miniSAR, where the left part represents the SAR images in the FARADSAR domain, the middle part represents the SAR images in the transition domain, and the right part represents the SAR images in miniSAR domain. (a)–(i) Column of SAR images. Among them, the image in the transition domain corresponds to the transform result of images in FARADSAR at the corresponding position.

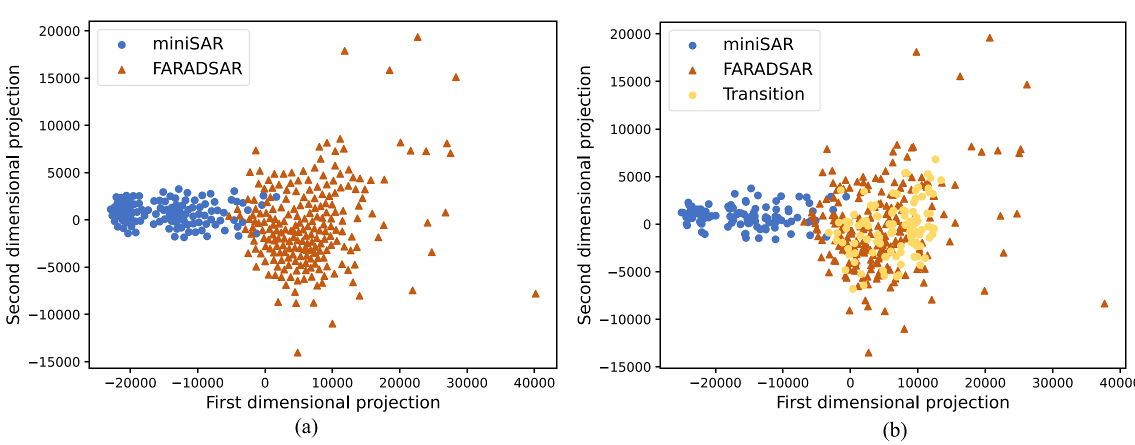

Fig. 9. Visualization of the original sample distributions via t-SNE. (a) and (b) Results of PDA from miniSAR (source domain) to FARADSAR (target domain), where each point represents an image. Blue: source images, Red: target images, orange: transform images.

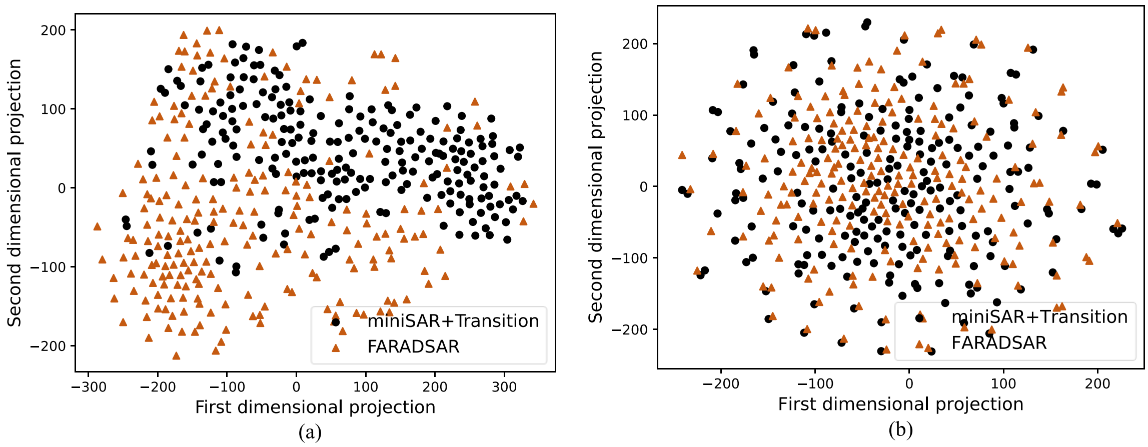

Fig. 10. Visualization of the feature distributions via t-SNE. (a) and (b) Results of MFDA from miniSAR and transition (diverse domain) to FARADSAR (target domain). For (a), each black dot represents one diverse image feature extracted from the feature extractor (last convolutional layer of the base network) before MFDA, and each red triangle represents one target image feature extracted from the same feature extractor. For (b), each black dot represents one diverse image feature extracted from the feature extractor (last convolutional layer of the base network) after MFDA, and each red triangle represents one target image feature extracted from the same feature extractor.

TABLE IV
Ablation Study on F2M

	PDA	MFDA	IPL	Precision	Recall	F1-score
Source only				0.9524	0.6504	0.7729
Ours	✓			0.9438	0.6829	0.7925
		✓		0.8911	0.7398	0.8084
			✓	0.8788	0.7154	0.7887
	✓	✓		0.9048	0.7805	0.8380
	✓		✓	0.8496	0.7724	0.8091
		✓	✓	0.8932	0.7561	0.8190
	✓	✓	✓	0.8595	0.8455	**0.8519**

TABLE V
Results of Reweighting in Different Levels for M2F Task

	Precision	Recall	F1-score
Low-level	0.7887	0.6831	**0.7316**
High-level	0.6160	0.7681	0.6837
No re-weighting	0.6912	0.7328	0.7108

and is closer to the target domain than the source domain, which verifies our motivation of PDA.

3) Analysis of MFDA: Fig. 10 visualizes feature distributions for M2F task by mapping the features to a low dimensional 2-D space. Fig. 10(a) is the distribution plot before MFDA, where diverse domain features and target domain features are obtained by the same feature extractor (last convolutional layer) before domain adaptation. This plot shows that the features of the two domains are relatively easy to distinguish. Fig. 10(b) shows the distribution plot after MFDA, where diverse domain features and target domain features are obtained by the same feature extractor after domain adaptation. From this plot, we can see that the feature distributions of two domains are more closely and difficult to distinguish after MFDA, which demonstrates that our MFDA can align feature distributions of the diverse domain and the target domain very well.

To demonstrate the effectiveness of our method, we give the results of reweighting at different levels for the M2F task in Table V, where low-level, high-level, no reweighting represented reweighting at low-level feature space (the seventh convolution layer of VGG16), reweighting at high-level feature space (the last convolution layer of VGG16), and unweighted operation, respectively. Reweighting at low-level feature space achieves the best result, which shows the effectiveness of our method. The result of reweighting at high-level feature space is even worse than unweighted. The reason may be that the similarity of SAR images is difficult to be measured by deeper features and reweighting at high-level feature space hurts the performance of the detector.

4) Analysis of IPL: To demonstrate the effectiveness of IPL, we set different experiments and evaluate the corresponding performance. Fig. 11 shows some pseudo-label results of IPL. It can be seen from the first row and second row of Fig. 11 that the samples selected for the first time with higher confidence average score have higher overall pseudo-label quality, whereas most of the remaining samples that are unselected with lower confidence average score have more false alarms and missed detections. The samples unselected for the first time with lower confidence average score do not participate in the first iteration training, which alleviates the negative impact of false pseudo-labels on network training. As shown in the third row of Fig. 11, after the first iteration training of pseudo-label samples with higher overall pseudo-label quality, the detection model further reduces the false alarms and missed detections of remaining samples, which improves the quality of pseudo-labels.

We also explored the impact of different selection thresholds and different situations on detection performance. Fig. 12

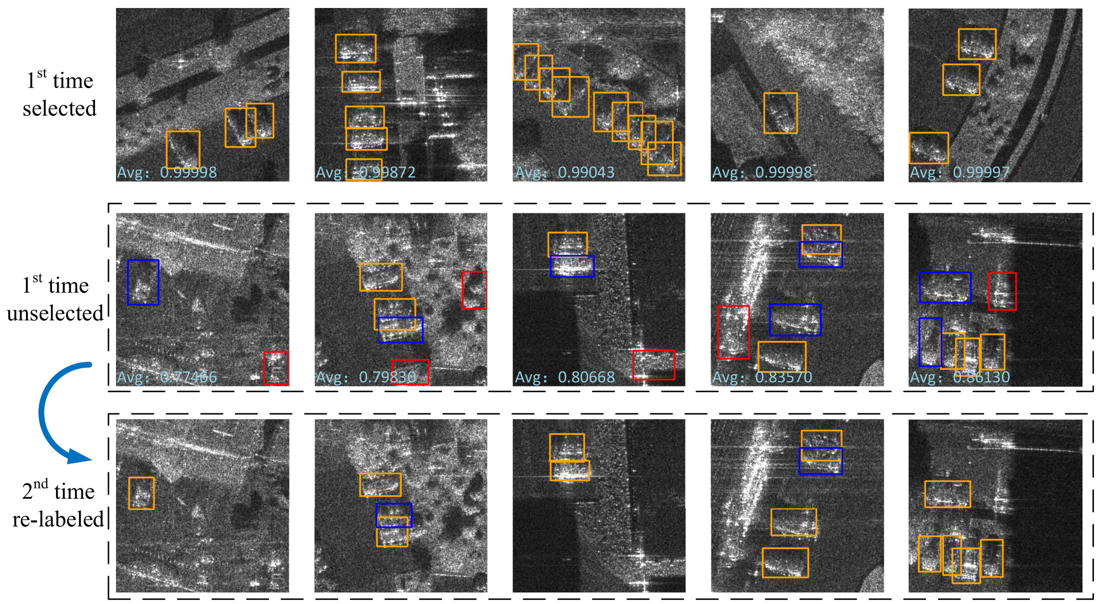

Fig. 11. Pseudolabel results for subimages in FARADSAR, where yellow rectangles represent the correctly detected target chips, and red rectangles represent false alarms and blue rectangles represent the missed detections. The first row represents some examples of the samples selected for the first time with a high average score. The second row represents some examples of the samples not selected for the first time with a low average score. The third row represents examples of the samples corresponding to the second row that are relabeled by the model after the first iteration of training.

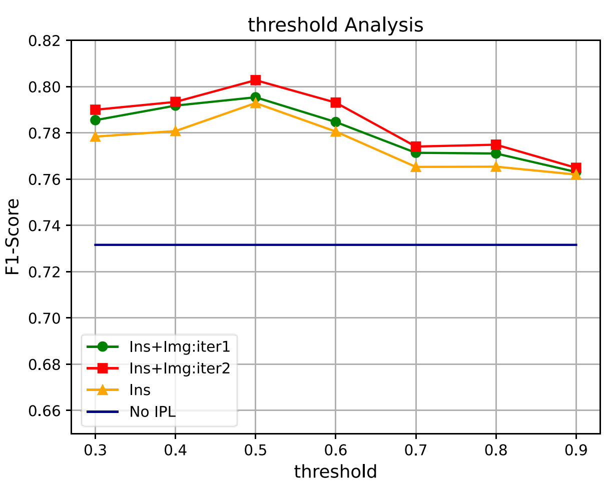

Fig. 12. Effect of thresholds with different experiment situations on detection performance.

shows the F1-score of four experimental situations with different thresholds, where “No IPL,”, “Ins,” “Ins+Img:iter1,” and “Ins+Img:iter2” denote the experiment without pseudo-labels, the experiment only using instance-level selected pseudo-labels, the first iterative experiment using instance-level and image-level selected pseudo-labels, and the second iterative experiment using instance-level and image-level selected pseudo-labels, respectively. It can be seen from this figure that the lower threshold (0.3–0.6) achieves better results than the threshold (0.7–0.9). This is because pseudo-label annotator constructed by MFDA and PDA produces relatively more missed detections. When the threshold is set to 0.5, the detector achieved the best performance. Compared with not using pseudo labeling strategy, since the more discriminative features of the target domain can be learned directly, the detection performance has achieved significant improvement by using pseudo labeling strategy. Compared with experiments only using instance-level selected pseudo-labels, the detection performance of the first iteration using both instance-level and image-level selected pseudo-labels has already achieved improvement, and the detector has achieved better detection performance at the second iteration. This is because our selection strategy alleviates the impact of error accumulation on network training effectively. When the selection threshold is set to 0.5, our method improves 7.12%, 1%, and 0.74% compared with “No IPL,” “Ins,” and “Ins+Img:iter1,” respectively. The compelling results clearly demonstrate that our IPL can learn more discriminative features in the target domain by IPL with instance-level and image-level selections.

5) Analysis of the Number of Labeled Source Domain Samples: Although the proposed method is unsupervised domain adaptation, the labeled SAR images in the source domain are required. As such, it is necessary to analyze the effect of the number of labeled SAR images in the source domain on the final detection performance. Fig. 13 gives the results of different percent of labeled source domain samples for the M2F task, where No IPL represents the result of our method without the IPL stage, IPL represents the final detection performance of our method, and baseline represents the result of using source images without domain adaptation. From this figure, we can see that the detection results are getting better with the increase of source domain labeled samples. The detector trained on the labeled source domain samples can be regarded as a teacher, and the target domain detector can be regarded as a student. The richer the labeled source domain samples, the more knowledge the source domain detector contains, and the better the performance of transferring to the target domain. But when the number of

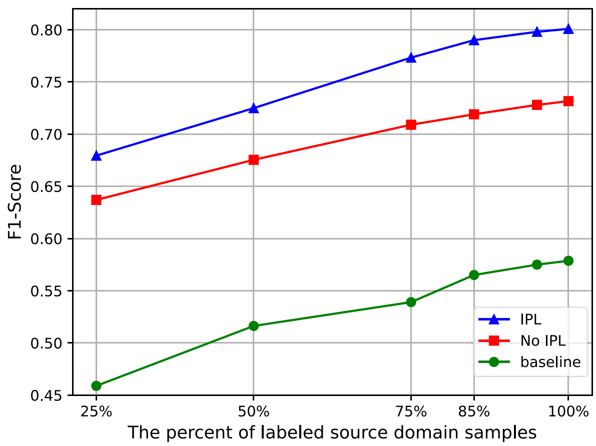

Fig. 13. Effect of the number of labeled source domain samples on detection performance.

labeled samples reaches 85% of the total samples, since the complete knowledge contained in the source domain can be roughly described, the performance improvement slows down and tends to remain unchanged.

IV. Conclusion

In this article, an unsupervised domain adaptation for SAR target detection is proposed. With our approach, the SAR target detection task is achieved in the target domain without any expensive label cost, which solves the problem of lacking labeled SAR images. The proposed method takes faster R-CNN as a detection network and constructs a three-stage domain adaptation framework for unsupervised learning. PDA and MFDA alleviate the performance drop by reducing the domain discrepancy. PDA can reduce the appearance differences through GANs and enhance the generalization ability of the model by enriching the distribution of the train data. MFDA can align the feature distribution at multilevel common feature space and reweight the low-level global features according to their relative importance to the target domain through adversarial learning. IPL is further proposed to solve the domain-biased problem by encouraging the detector to learn more discriminative features of the target domain directly. The experimental results based on the measured miniSAR dataset and FARADSAR dataset outperform the baseline and other comparison methods by a clear margin, which demonstrates the effectiveness of our method. In the future, making full use of special imaging mechanism characteristics of the SAR images for detection is also worthy of follow-up research.

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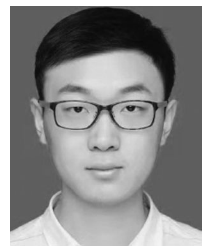

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